A SIDE PROJECT · THREE.JS · AUTOMATED QA

I built a robot that plays my game to find its bugs.



The Vanishing Gallery is a hand-drawn art-heist game on a tiny planet. It ships to the web on **every push**. So I gave it a test harness.

swipe →

THE PROBLEM

Every push goes live. Every bug, too.

A browser game: ~15,000 lines of plain JavaScript, no build step, deployed by Netlify on every commit. My feature log kept filling up with the same kinds of bug:

- planters floating over the lawn
- feet sunk into the floor
- a road through a building
- a hero who moonwalks
- a ladder hanging in mid-air
- a plant in the middle of the street

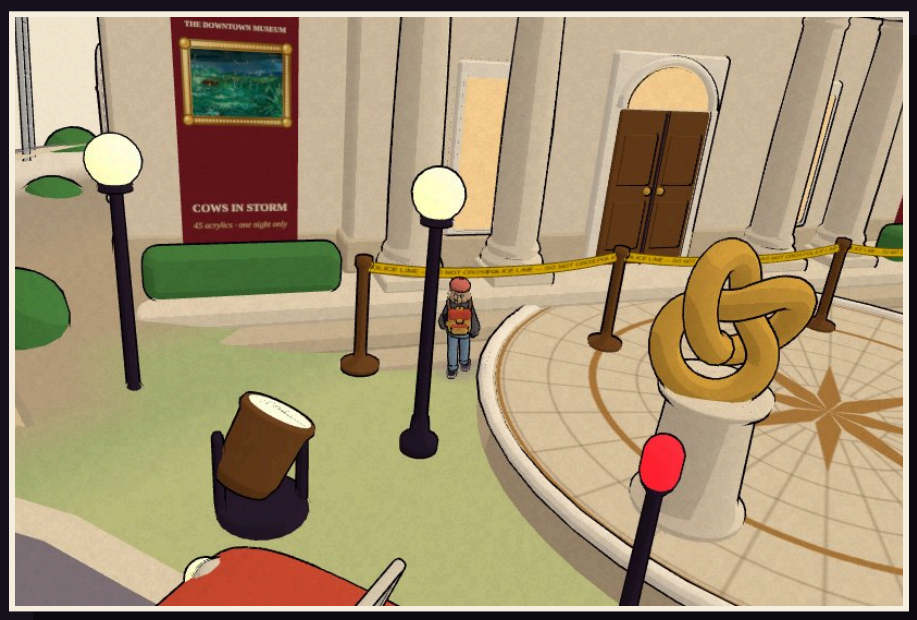
~220 lessons and rules, written down in my own docs, that nothing was checking.

HOW IT WORKS

It plays the real game, like a person would.

1. Boots the actual game in headless Chrome (Playwright)
2. Presses the keys and clicks a player would: walk, talk, drive, accuse
3. Reads the live 3D scene and measures it against my rules
4. Every failure says what, where and which rule — with a photo





Screenshots the harness took mid-run: a conversation, and the car on the road.

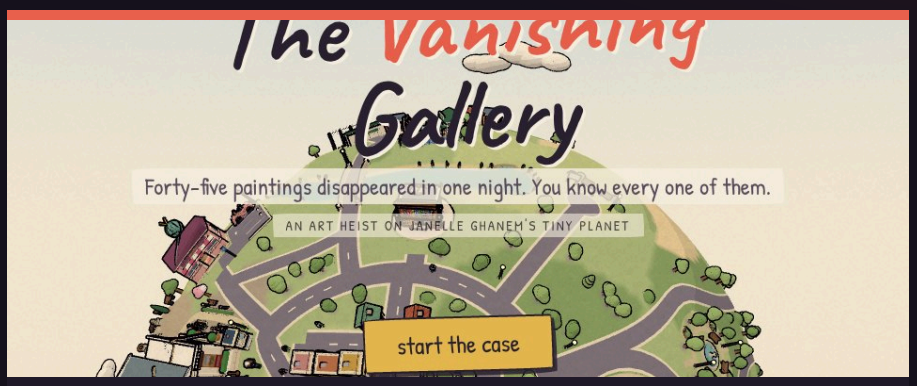
WHAT IT CHECKS

Six suites. 65 tests.



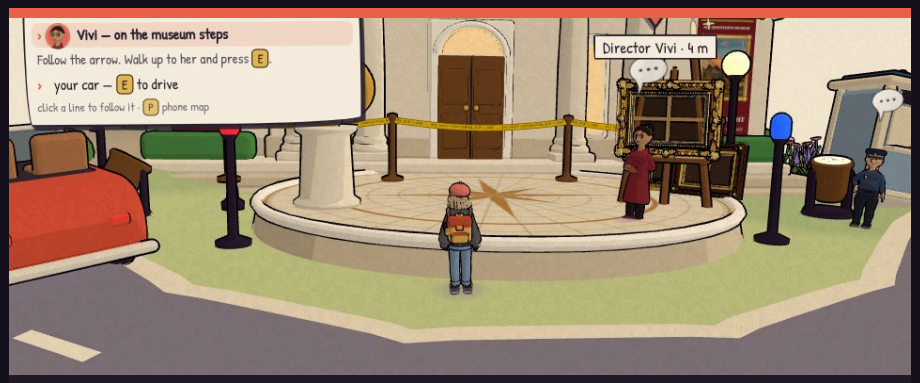
static

imports, cache tokens, asset budgets — 2 seconds



story

title to "Case closed", every clue, every suspect



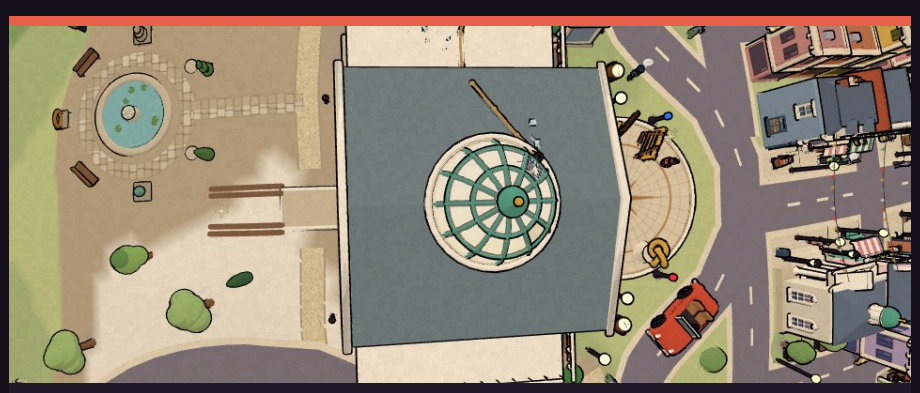
interactions

every key, panel, door, car and boat



models

broken geometry, scale, triangle budgets



rules

nothing floats, roads round buildings, no emoji...



rigs

skeletons, posture, gait — no moonwalking

BUG #1 · REAL, NOW FIXED

The swimmer was standing in the pool.

BEFORE



AFTER

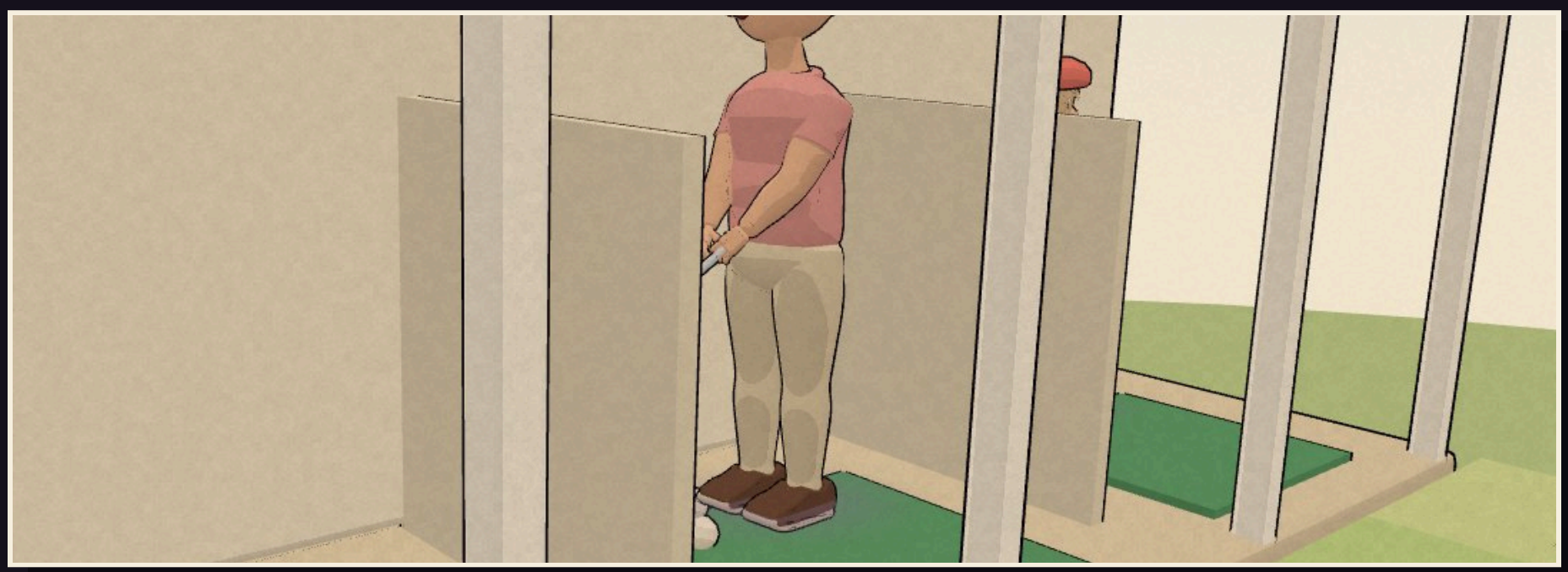


I spotted this one by eye. The harness found *why*: she was a regular townsperson in the walking pose, sunk to her chest. Now she swims breaststroke, and a test fails if she ever stands up again.

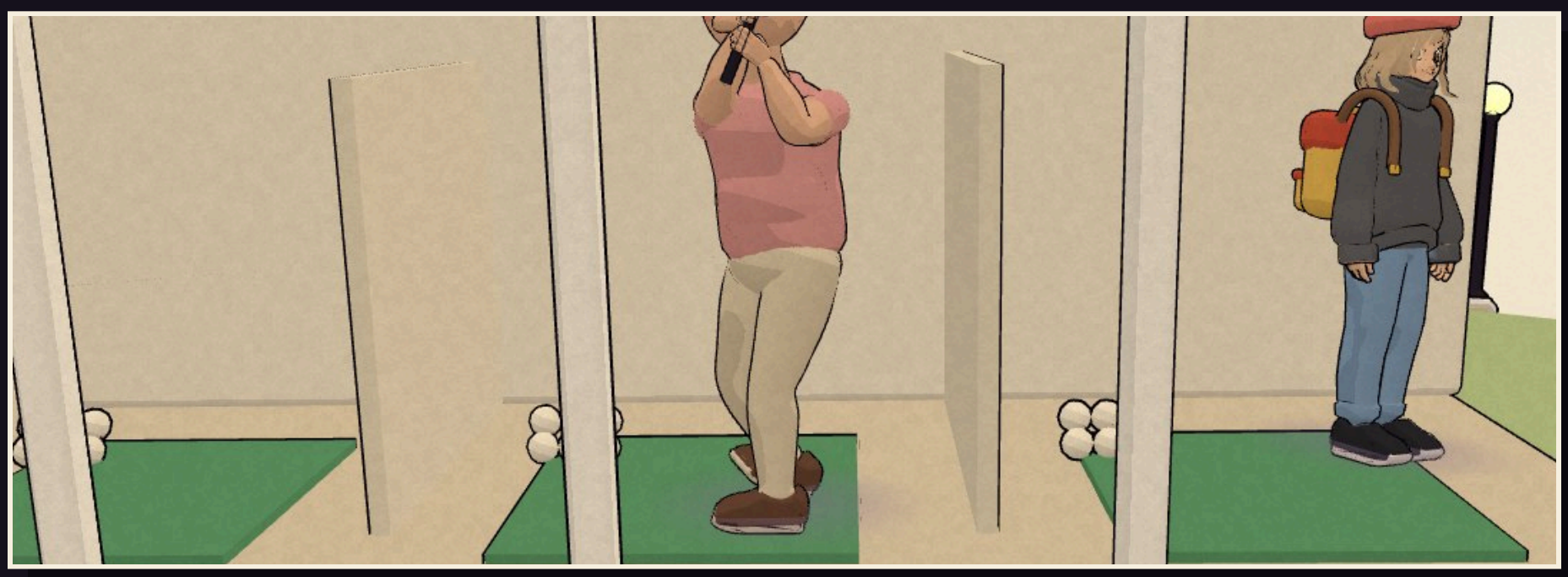
BUG #2 · REAL, NOW FIXED

The golfer floated and never swung.

BEFORE



AFTER



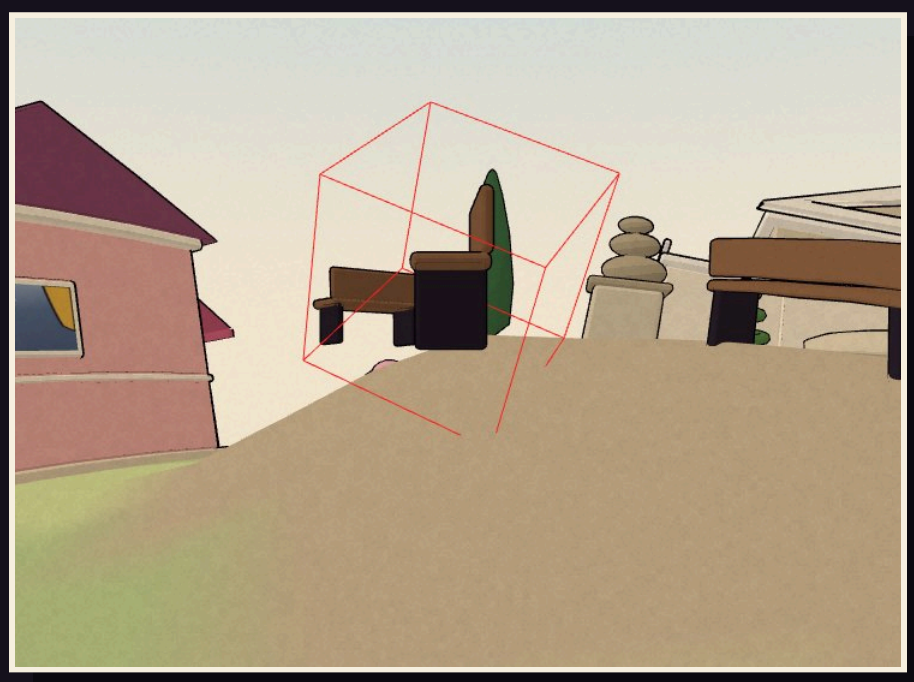
```
✖ fists 0.141 apart — not both on the grip
✖ ankles never below 0.126 over the mat   (0.034 fixed)
```

A REAL TEST RUN

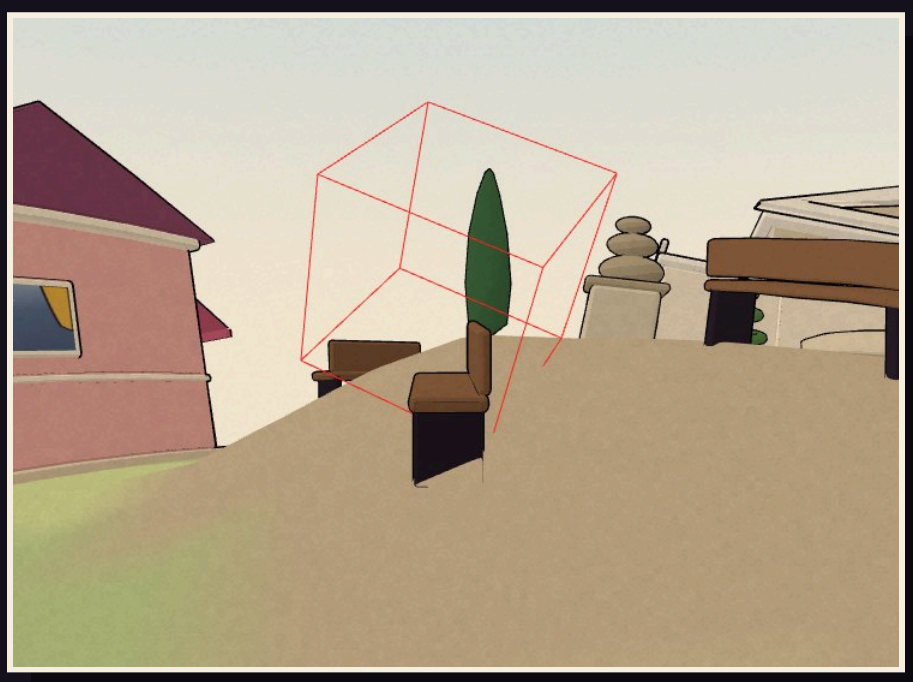
I added two benches.
It caught two mistakes.

```
$ node tests/run.mjs static
✖ modules missing from the import map: 'benches.js'
$ node tests/run.mjs rules
✖ nothing floats  gap: 0.281  near: 'museum-garden'
ℹ photo of new floater: floating-1.png
```

CAUGHT



FIXED · 16/16 GREEN



Neither shows on the title screen or a quick walk past. Both would have shipped.

POINTED AT THE LIVE SITE

After 3 big merges, it went red.

Same tests, unchanged, run against what's on the web today:

```
✓ static        ✓ interactions        ✓ rigs
✗ story         ✗ models              ✗ rules
```

A clue lost its painted sketch
a real regression: the ladder card would show with no picture

Shader programs 53 → 68
a performance budget question

Five new things off the ground
each one photographed: a bridge (fine), two to check up close

THE PART NOBODY POSTS

The harness was wrong six times first.

> **"The hero is 8.6 units tall"**
> her model reshapes on the GPU — now measured by her skeleton

> **"Five things are floating!"**
> it was the bunting over Main Street — meant to hang there

> **"Everyone sank 0.4"**
> hidden triangles under clothes — now skipped

> **"She moonwalks 98% of the time"**
> wrong metric — now uses the one from my own bug log

Lesson: **look at a screenshot before you trust a number.** Every exception is checked in a picture and written down with a reason.

BY THE NUMBERS

Small game. Real QA.

6
test suites

65
tests

~220
rules catalogued from my own docs

0
changes to how the game is built

2
real bugs fixed, with tests that guard them

~15 min
for the whole run, in software rendering

Built on ISTQB's game-testing syllabus and the way studios like Rare test *Sea of Thieves* — scaled down to one person and a browser.

What would you test first?

github.com/ghanemja/3js_test_harness · art-heist.netlify.app